Automated Construction of Multi-Element MEAM Interatomic Potentials

DFT-Informed Neural Network Optimization

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PHYS 400 — Mock Presentation

The Problem

- Industrial alloys (Al 7075, steels) are chosen by their **elastic properties** — yet measuring them experimentally is **slow and expensive**.
- Molecular dynamics can predict these properties *computationally*, but only with accurate **interatomic potentials**.
- The Modified Embedded Atom Method (MEAM) is the standard for metallic alloys.

Goal

Build an automated pipeline that generates MEAM potentials for *arbitrary* multi-element alloys.

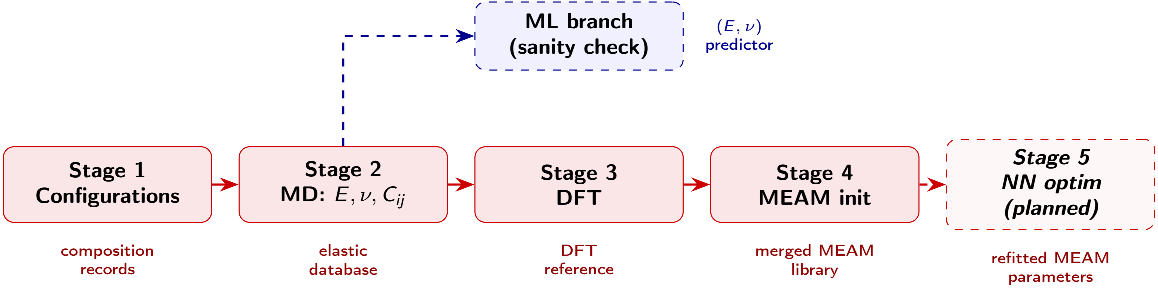
Why Existing Approaches Fall Short

Published MEAM files cover only **subsets** of elements. No single file spans Al 7075 (Al, Zn, Mg, Cu, Cr, Mn, Fe, Si, Ti).

Library File	Al	Co	Cr	Cu	Fe	Mg	Mn	Mo	Ni	Si	Ti	Zn
AlMgZn	✓					✓						✓
AlSiCuMgFe	✓			✓	✓	✓				✓		
CuMo				✓				✓				
FeMnNiTiCuCrCoAl	✓	✓	✓	✓	✓		✓		✓		✓	

- **Cross-interaction parameters** between files are undefined $\Rightarrow$ cannot simulate combined compositions.
- **Pure ML potentials** (MLIPs) are accurate but demand massive datasets and sacrifice MEAM's transparency.

Pipeline Overview



Pipeline Stages at a Glance

Stage	Description	Framework	Status
1	Random alloy configuration generation	Python	Complete
2	MD elastic-property evaluation	LAMMPS Python API	Complete
3	DFT reference calculations	Quantum Espresso + ASE	Complete
4	MEAM initialization & library merging	Python	Complete
5	Neural-network surrogate optimization	TensorFlow + LAMMPS	*Planned*

Key Idea

Use neural networks as a **surrogate** that learns the MEAM-parameters $\rightarrow$ elastic-properties map, then inverse-optimize parameters to hit target (E, ν).